
Stochastic Actor-Critic: Mitigating Overestimation Bias with a Single Distributional Critic

Uğurcan Özalp

Turkish Aerospace
Ankara, Turkey

ugurcanozalp06@gmail.com

Abstract

Off-policy actor-critic methods in reinforcement learning train a critic with temporal-difference updates and use it as a learning signal for the policy (actor). This design typically achieves higher sample efficiency than purely on-policy methods. However, critic networks tend to overestimate value estimates systematically. This is often addressed by introducing a pessimistic bias based on uncertainty estimates. Current methods employ ensembling to quantify the critic’s epistemic uncertainty (arising from limited data) to scale pessimistic updates. In this work, we propose a new algorithm called Stochastic Actor-Critic (STAC) that incorporates temporal (one-step) aleatoric uncertainty (due to noise in the data) of the critic to scale pessimistic bias in the temporal-difference targets, rather than relying on epistemic uncertainty. STAC uses a single distributional critic network to model the temporal distribution of return, and applies dropout to both the critic and actor networks for regularization. Our results show that pessimism based on a distributional critic alone suffices to mitigate overestimation, and leads to risk-averse behavior in stochastic environments. Introducing dropout further improves training stability and performance by means of regularization. With this combination, STAC achieves better computational efficiency using a single distributional critic network.

1 Introduction

Actor-critic methods leverage off-policy samples to train critics, promising higher sample-efficient learning than on-policy algorithms. Despite this advantage, actor-critic agents often struggle with *overestimation bias* in value function learning due to the joint effects of function approximation, temporal-difference learning, and off-policy sampling (Thrun & Schwartz, 2014; Sutton & Barto, 2018; Van Hasselt et al., 2018), which can destabilize training. In this work, we revisit this problem from a fresh perspective, focusing on the use of uncertainty modeling in actor-critic architectures.

Epistemic uncertainty indicates a lack of training data and is used by current actor-critic methods to scale pessimistic updates (Fujimoto et al., 2018; Haarnoja et al., 2018; Kuznetsov et al., 2020; Moskovitz et al., 2021; Chen et al., 2021; Hiraoka et al., 2021; Zhang et al., 2024). These methods use critic ensembles for this kind of uncertainty modeling. However, pessimistic updates based on epistemic uncertainty in the critic may hinder exploration of the state-action space, contradicting the principle of *optimism in the face of uncertainty* (Kocsis & Szepesvári, 2006; Audibert et al., 2007; Azizzadenesheli et al., 2018; Ciosek et al., 2019; O’Donoghue, 2023; Wu et al., 2023).

On the other hand, aleatoric uncertainty is a measure of noise within training data. Some works propose to use pessimistic updates based on both epistemic and aleatoric uncertainty of return (discounted cumulative reward) for overestimation mitigation (Kuznetsov et al., 2020). Originally, modeling aleatoric uncertainty is a topic of distributional reinforcement learning (Bellemare et al., 2017; Dabney et al., 2018b; Kim & Oh, 2022; Duan et al., 2021; 2025; Ma et al., 2025), and mainly used to scale risk sensitivity of the agent (Tang et al., 2019; Yang et al., 2021; Théate & Ernst, 2023; Ma et al., 2025). While pessimistic policy updates lead to risk-averse behavior, optimistic updates result in risk-seeking behavior. Recently, in Q-learning setting,

Achab et al. (2023) proposed to model temporal aleatoric distribution of return, in which uncertainty is induced by only one-step dynamics of the environment, rather than across all steps. This approach is proven to have convergence guarantees theoretically.

In off-policy actor-critic methods, the temporal-difference target changes continuously as the policy is updated. Although policy improvement is deterministic given a critic, high-dimensional function approximation causes small critic errors to be selectively exploited in a non-stationary manner, inducing irreducible variability in value targets even in deterministic environments, which is a major source of overestimation. We argue that this variability is best captured as temporal aleatoric uncertainty in the critic output, as it aggregates all uncertainty sources contributing to overestimation: policy-induced effects and transition or reward stochasticity.

In this work, we show that temporal aleatoric uncertainty can be modeled by a distributional critic, and that pessimism applied to this uncertainty alone is sufficient to control overestimation without ensembles or multiple critics. Based on this idea, we introduce a novel off-policy distributional actor-critic algorithm, Stochastic Actor-Critic (STAC), specifically modeling one-step return uncertainty for critic, and pessimistic updates based on this distribution. This way, STAC eliminates the need for double/ensemble critics, reducing both computation and memory costs. In other words, STAC uses *pessimism in the face of temporal aleatoric uncertainty* for overestimation mitigation.

Our method differs from other methods using aleatoric uncertainty for risk aversion Tang et al. (2019); Yang et al. (2021); Kim & Oh (2022); Ma et al. (2025). These methods, incorporate pessimism on total return uncertainty only for actor updates, whereas STAC uses pessimism on temporal uncertainty for both critic and actor updates. The main purpose is overestimation mitigation rather than risk-averse learning, but pessimism in STAC still yields to risk aversion, similar to fully distributional methods.

Recent work of Nauman et al. (2024) has also shown that network regularization has positive impact on overestimation by reducing overfitting. Ensemble based methods inherently regularized by using multiple critic networks. Since STAC uses a single critic network, it is sensitive to learning errors and overfitting. For this purpose, dropout (Srivastava et al., 2014) and layer normalization (Ba et al., 2016) are employed in both actor and critic networks as regularization tools.

The implementation is very simple and can be obtained by introducing dropout to networks, modeling a distributional critic and defining a pessimistic learning objective upon Soft Actor-Critic algorithm (Haarnoja et al., 2018). We conduct experiments on standard RL benchmarks to evaluate the performance of STAC compared to existing methods. Our results demonstrate the effectiveness of STAC in achieving competitive performance with state-of-the-art methods while requiring fewer computational resources (single critic), making it a promising approach for real-world RL applications.

2 Preliminaries

This section introduces the minimum background required to analyze uncertainty-induced overestimation in off-policy actor-critic methods. Throughout the paper, $\mathcal{P}(\Omega)$ denotes the set of all possible probability distributions on the set Ω . $A \stackrel{D}{=} B$ indicates that two random variables, A and B , have identical probability laws.

2.1 Aleatoric and Epistemic Uncertainty

Uncertainty in an estimate is mainly categorized as either *aleatoric* or *epistemic* (Der Kiureghian & Ditlevsen, 2009; Kendall & Gal, 2017; Gal et al., 2016b). Epistemic uncertainty refers to uncertainty of model parameters (θ) due to insufficient training data. Given the training data $\mathcal{D}$ and prior distribution $p(\theta)$ over the network parameters θ , we can compute the posterior distribution over the parameters $p(\theta | \mathcal{D})$ using Bayesian inference.

On the other hand, aleatoric uncertainty is induced by the inherent randomness within the data. Even with infinite data, it is unavoidable and cannot be reduced, because it is an intrinsic part of the process being modeled. In deep learning, we can model this by having a distributional network that outputs a probability distribution $p_\theta(y|x)$ conditioned on input x (Lakshminarayanan et al., 2017; Kendall & Gal, 2017). Given

a dataset $\mathcal{D}$, the loss function for training the network can be derived from the negative log-likelihood: $\mathcal{L}_\theta(\mathcal{D}) = \mathbb{E}_{\{(x_i, y_i)\} \sim \mathcal{D}} [-\log p_\theta(y_i | x_i)]$. In reinforcement learning, aleatoric uncertainty naturally arises from stochastic rewards, transition dynamics, and policy-induced randomness in temporal-difference targets.

2.2 Maximum Entropy Actor-Critic Reinforcement Learning

In reinforcement learning language, the agent lives in a Markov Decision Process (MDP) which is represented by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, d_0, \tau, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $d_0 \in \mathcal{P}(\mathcal{S})$ is the initial state distribution, $\tau : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$ is the transition kernel and $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function.

The initial state is sampled first, $s_0 \sim d_0(\cdot)$. At time step t , being in state s_t ; next state is obtained from the environment, $s_{t+1} \sim \tau(\cdot | s_t, a_t)$ depending on the taken action $a_t \sim \pi(\cdot | s_t)$. Finally, a reward is obtained, $r_t = R(s_t, a_t)$ from the reward function R . Return and action return are defined respectively as follows,

$$G^\pi(s) \stackrel{\text{D}}{=} \sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) - \alpha \log \pi(a_t | s_t)), \quad s_0 = s, \quad (1)$$

$$Z^\pi(s, a) \stackrel{\text{D}}{=} R(s, a) + \sum_{t=1}^{\infty} \gamma^t (R(s_t, a_t) - \alpha \log \pi(a_t | s_t)), \quad s_0 = s, \quad a_0 = a, \quad (2)$$

The ultimate goal of the agent is to derive a policy $\pi : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{A})$ to maximize discounted return G^π with entropy bonus (Haarnoja et al., 2017; 2018). However, returns are random variables, and cannot be used as objective. Therefore, value (V) and action-value (Q) functions are defined as expectations of return over policy and transition dynamics, $V^\pi(s) = \mathbb{E}_{\pi, \tau} [G^\pi(s)]$, $Q^\pi(s, a) = \mathbb{E}_{\pi, \tau} [Z^\pi(s, a)]$. Actor-critic methods model action value function Q , and learning iterates between solving policy evaluation and policy improvement.

Policy evaluation minimizes the temporal difference: $\|\mathcal{T}^\pi Q(s, a) - Q(s, a)\|$ by a gradient step, where $\mathcal{T}^\pi Q$ is the Bellman backup operator, and defined as,

$$\mathcal{T}^\pi Q(s, a) = R(s, a) + \gamma \mathbb{E}_{\substack{s' \sim \tau(\cdot | s, a) \\ a' \sim \pi(\cdot | s')}} [Q(s', a') - \alpha \log \pi(a' | s')]. \quad (3)$$

Policy improvement is minimizing value function $V^\pi(s) = \mathbb{E}_{a \sim \pi} [Q^\pi(s, a) - \alpha \log \pi(a | s)]$ by a gradient step. After policy improvement and policy evaluation at step k , updated action value function at step $k+1$ is $Q^{k+1}(s, a) = \mathcal{T}^* Q^k(s, a)$, where $\mathcal{T}^*$ is the Bellman optimality operator (Equation 5 from Haarnoja et al. (2017)),

$$\mathcal{T}^* Q(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim \tau(\cdot | s, a)} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right]. \quad (4)$$

$\tilde{\alpha} \in [\alpha, \infty)$ is defined to demonstrate the continuum of policy improvement, i.e., the fact that policy is updated by partially exploiting state-action value function at each iteration. As $\tilde{\alpha} \rightarrow \infty$, $\mathcal{T}^* Q$ boils down to $\mathcal{T}^u Q$, which represents action value for uniform policy. On the other hand, $\tilde{\alpha} = \alpha$ represents updated action value after complete exploitation. Consequently, in deep learning setting, a higher learning rate of actor network results in lower $\tilde{\alpha}$, closer to α .

3 Distributional Reinforcement Learning and Overestimation

3.1 Distributional Maximum Entropy Actor-Critic

Instead of learning action value function Q , action return can also be modeled as a random variable to identify possible consequences of a given policy (Bellemare et al., 2017). This also constitutes the methodology of distributional maximum entropy actor-critic methods (Duan et al., 2021; Ma et al., 2025). For this,

action return distribution $\mathcal{Z} \in \mathcal{P}(\mathbb{R}^{S \times \mathcal{A}})$ is defined, where action returns are sampled from this distribution, $Z(s, a) \sim \mathcal{Z}(s, a)$. Recall that $Q(s, a) = \mathbb{E}_{\mathcal{Z}, \tau}[Z(s, a)]$. The corresponding Bellman backup operator $\mathcal{T}^\pi$ is defined as,

$$\mathcal{T}^\pi Z(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma(Z(s', a') - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'), \quad (5)$$

To make notation easier, distributional Bellman backup $\mathcal{T}_D^\pi \mathcal{Z}$ is defined, where Bellman backup of sampled values are sampled from this distribution, i.e., $\mathcal{T}^\pi Z(s, a) \sim \mathcal{T}_D^\pi \mathcal{Z}(s, a)$. At each step k , the state-action return distribution is updated by minimizing Kullback-Leibler divergence from distributional Bellman backup: $D_{\text{KL}}(\mathcal{T}_D^\pi \mathcal{Z}(s, a) || \mathcal{Z}(s, a))$ by a gradient step.

One-step Distributional Uncertainty Unlike conventional distributional modeling, it is also possible to model action return with randomness only up to first step Achab (2020); Achab et al. (2023). In this approach, Bellman backup uses expectation (deterministic) of return of next state and action. This way, uncertainty due to environment stochasticity and actor-induced uncertainty (due to policy improvement) are modeled for only one step, which directly targets the source of critic overestimation. This distribution is denoted as $\mathcal{Q}$ instead of $\mathcal{Z}$. Therefore, $Q \sim \mathcal{Q}$ is a random variable unlike previous formulation. Bellman backup is defined similar as in Equation 5 but expectation of next-state value is used,

$$\mathcal{T}^\pi Q(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma(\mathbb{E}_{Q \sim \mathcal{Q}}[Q(s', a')] - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'). \quad (6)$$

At each step k , the state-action value distribution is updated by minimizing Kullback-Leibler divergence from distributional Bellman backup: $D_{\text{KL}}(\mathcal{T}_D^\pi Q(s, a) || Q(s, a))$ by a gradient step. Corresponding Bellman optimality operator is as follows,

$$\mathcal{T}^* Q(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right], \quad s' \sim \tau(\cdot | s, a). \quad (7)$$

This operator is proved to be a contraction in Q-learning by Achab et al. (2023), using max instead of logsumexp operator. This guarantees learning convergence in tabular setting. However, in function approximation, there are other phenomena like overestimation and overfitting. As a main focus of this work, we will discuss overestimation bounds of one-step Bellman optimality operator, and propose a method to mitigate it in the next section.

3.2 Quantifying Overestimation for Sub-Gaussian Critic Distributions

Let $\mu(s, a) = \mathbb{E}_{\mathcal{Q}, \tau}[Q(s, a)]$, the Bellman optimality backup of the mean is not equal to the mean of the backup, $\mathcal{T}^* \mu(s, a) \neq \mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)]$ due to policy improvement, and the difference is the overestimation bias. In this section, we analyze the overestimation bias similar to Chen et al. (2021) and Lan et al. (2020) but in the soft learning framework instead of discrete actions. We define the overestimation error as the difference between $\mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)]$ and average $\mathcal{T}^* \mu(s, a)$ as ϵ ,

$$\epsilon(s, a) = \mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)] - \mathcal{T}^* \mu(s, a). \quad (8)$$

In the ideal case, $\epsilon(s, a)$ should be zero if there is no overestimation, which is not the case due to critic uncertainty. To quantify it, we assume that critic distribution $\mathcal{Q}(s, a)$ is sub-Gaussian with variance proxy $\sigma^2(s, a)$ and mean $\mu(s, a)$. We adopt a sub-Gaussian assumption as it provides a mild and widely used concentration model that enables analytic overestimation bounds without restricting the critic to a specific parametric distribution.

Definition 3.1. A random variable $X \in \mathbb{R}$ with mean $\mu = \mathbb{E}[X]$ is called sub-Gaussian with variance proxy σ^2 if its moment generating function satisfies

$$\mathbb{E}[\exp(\lambda X)] \leq \exp \left(\lambda \mu + \frac{1}{2} \lambda^2 \sigma^2 \right), \quad \forall \lambda \in \mathbb{R}. \quad (9)$$

Under the sub-Gaussian assumption, Theorem 3.1 quantifies how critic uncertainty propagates through the soft Bellman optimality operator and Corollary 3.1.1 motivates a variance-dependent pessimistic correction.

Theorem 3.1 (Overestimation quantification for sub-Gaussian critics). *Let given state-action value distribution $\mathcal{Q} \in \mathcal{P}(\mathbb{R}^{S \times A})$ be sub-Gaussian with mean $\mu(s, a)$ and variance proxy $\sigma^2(s, a)$ for all state-action pairs, with bounded support. For each sample $Q \sim \mathcal{Q}$,*

$$\mathcal{T}^*Q(s, a) \leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_A \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right). \quad (10)$$

Corollary 3.1.1 (Pessimistic critic target). *For policy evaluation, using pessimistic Bellman backup,*

$$\mathcal{T}_\beta^\pi Q(s, a) \stackrel{D}{=} R(s, a) + \gamma (\mu(s', a') - \beta \sigma(s', a') - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'), \quad (11)$$

prevents overestimation as long as $\beta \geq \max_{(s, a)} \frac{1}{2} \tilde{\alpha}^{-1} \sigma(s, a)$.

Moreover, maximum overestimation bound is demonstrated in Theorem 3.2.

Theorem 3.2 (Overestimation bound). *Overestimation due to uncertainty of distribution $\mathcal{Q}$, denoted as ϵ , is upper bounded for Bellman updates,*

$$\epsilon(s, a) \leq \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right]. \quad (12)$$

As an additional notation, we introduce pessimistic distributional Bellman backup $\mathcal{T}_{\beta, D}^\pi \mathcal{Q}$ where pessimistic Bellman backup of sampled values are sampled from this distribution, i.e., $\mathcal{T}_\beta^\pi Q(s, a) \sim \mathcal{T}_{\beta, D}^\pi \mathcal{Q}(s, a)$. The difference between $\mathcal{T}_{\beta, D}^\pi \mathcal{Q}(s, a)$ and $\mathcal{T}_D^\pi \mathcal{Q}(s, a)$ is illustrated in Figure 1.

Taken together, these results show that overestimation in off-policy actor-critic methods originates from temporal uncertainty in Bellman targets, and that a simple variance-proxy-based pessimistic shift is sufficient to control it. This observation motivates the Stochastic Actor-Critic algorithm introduced next.

4 Stochastic Actor-Critic

In this section, we discuss key mechanisms needed for computation and sample efficient actor-critic learning and propose our algorithm *Stochastic Actor-Critic*. This algorithm employs a *single distributional* critic network $\mathcal{Q}_\theta$ by parameter set θ which captures temporal (one-step) aleatoric uncertainty, and dropout along with layer normalization for network regularization. Actor network π_ϕ outputs a **tanh** transformed normal distribution to bound actions to $[-1, 1]$, and is parameterized by set ϕ . Unlike other methods, STAC uses critic prediction in a pessimistic manner using temporal aleatoric uncertainty, for policy evaluation and policy improvement.

The rationale behind this argument is that most of the temporal difference target randomness due to environment stochasticity and ongoing actor updates (policy improvement) inherently appears in the form of aleatoric uncertainty from the critic’s side, and overestimation is the result of this randomness. Although policy improvement itself is deterministic given a critic, small approximation errors in the critic are selectively exploited by the policy update in a non-stationary manner, which induces unpredictable variability in Bellman targets in practice. Unlike distributional methods (Bellemare et al., 2017), we do not model aleatoric uncertainty of the full return, since it is not directly related to overestimation.

Although return uncertainty is modeled only for one-step, pessimism in STAC also leads to risk-averse learning since critic is also learned by pessimistic updates. Conventional distributional algorithms model full return as a distribution. For risk-averse learning only actor is updated in pessimistic way, not critic, usually by conditional value at risk measure (CVaR) (Dabney et al., 2018a; Tang et al., 2019; Yang et al., 2021; Kim & Oh, 2022; Ma et al., 2025) or by standard deviation (Ma et al., 2025).

Epistemic pessimism naturally decreases the degree of overestimation by decreasing probability of critic errors which are exploited by policy improvement later. However, out-of-distribution states and actions are

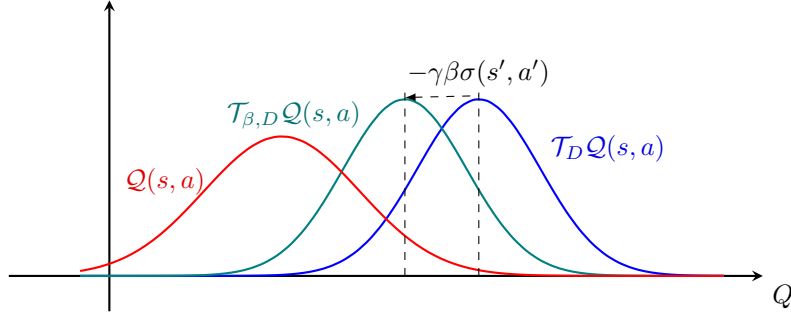


Figure 1: Evolution of the Pessimistic Distributional Bellman Backup. While $\mathcal{T}_D^\pi Q(s, a)$ is overestimated, a corrected backup $\mathcal{T}_{\beta, D}^\pi Q(s, a)$ is closer to $Q(s, a)$.

detected by high epistemic uncertainty, and should not be suppressed by pessimism. On the contrary, these state-action pairs should be visited for exploration. Therefore, epistemic uncertainty should not be used as a pessimism signal for overestimation mitigation unlike previous methods.

(Nauman et al., 2024) demonstrated that network regularization methods improve performance. Since STAC employs a single critic network, it is prone to overfitting more than ensemble based methods. To prevent it, STAC employs dropout (Srivastava et al., 2014) similar to Hiraoka et al. (2021). While dropout may also promote exploration by injecting stochasticity into the policy and value estimates, we do not attempt to isolate or prove this effect here. Dropout also allows capturing the probabilistic nature of a network, representing Bayesian neural networks (Gal & Ghahramani, 2016). However, it is important to note that STAC employs dropout on both critic and actor networks only for regularization purposes, not for epistemic modeling.

4.1 Policy Evaluation

For simplicity, STAC models critic as normal distribution. This contradicts the bounded distribution assumption of Theorem 3.1, but it yields a simple loss function and is easy to interpret. Still, it is reasonable to assume critic distribution is bounded for finite horizon or discounted MDPs ($\gamma < 1$) with bounded reward functions.

Using transition tuples from experience replay as batch, $\mathcal{D}_b = \{(s_i, a_i, r_i, s'_i, \text{done}_i)\}_{i=1}^{N_b}$, temporal difference (TD) target Q_i^{TD} , representing Bellman backup, is β -pessimistic,

$$Q_i^{TD} = r_i + \gamma(\mu_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \beta\sigma_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \alpha \log \pi_{\phi}(s'_i, \tilde{a}'_i))(\neg \text{done}_i), \quad \tilde{a}'_i \sim \pi_{\phi}(\cdot | s'_i). \quad (13)$$

Learning objective is cross-entropy loss (log loss),

$$\mathcal{L}_{\theta}(\mathcal{D}_b) = \frac{1}{N_b} \sum_{i=1}^{N_b} -\log \mathcal{Q}_{\theta}(Q_i^{TD} | s_i, a_i) = \frac{1}{2} \log 2\pi + \frac{1}{2N_b} \sum_{i=1}^{N_b} \left(\log \sigma_{\theta}^2(s_i, a_i) + \frac{(Q_i^{TD} - \mu_{\theta}(s_i, a_i))^2}{\sigma_{\theta}^2(s_i, a_i)} \right). \quad (14)$$

4.2 Policy Improvement

According to Corollary 3.1.1, pessimistic TD targets should be used to train critic network. However, this analysis does not account for policy learning since policy is assumed as softmax over action values. In actor-critic framework, same pessimistic objective should be used for policy improvement, since policy evaluation and policy improvement objective should be the same. In other words, at learning step k , Bellman backup must be equal to Bellman policy evaluation with the new policy, $\mathcal{T}^* Q^k \stackrel{D}{=} \mathcal{T}^{\pi^{k+1}} Q^k$. It is only possible by using same pessimistic critic value for policy improvement.

4.3 Other Details and Algorithm Summary

Layer Normalization We implement Layer Normalization (Ba et al., 2016) after all hidden activations of critic and actor networks, in order to stabilize learning and prevent possible numerical instabilities.

Lagged critic for TD target When the trained critic network is also used in calculating the target value, the critic training is prone to divergence (Li et al., 2023). For this, a common approach is to use another critic network to evaluate TD target (Mnih et al., 2013). Similar to Lillicrap et al. (2015), Fujimoto et al. (2018), and Haarnoja et al. (2018), we use a delayed form of critic network for TD target evaluations as demonstrated in Equation 13. The parameters of target critic are only updated by Polyak averaging of main critic network weights through learning steps; $\bar{\theta} \leftarrow \rho\bar{\theta} + (1 - \rho)\theta$. This strategy is important to ensure the stability of temporal difference learning.

Automatic temperature tuning Using constant temperature results in different policies if the reward magnitude changes. To mitigate this, Haarnoja et al. (2018) proposed a policy entropy constraint, representing temperature as the Lagrange multiplier of the constraint. Given target entropy $\bar{\mathcal{H}}$ as hyper-parameter, the loss function related to this constraint is as follows;

$$\mathcal{L}_\alpha(\mathcal{D}_b) = -\alpha\bar{\mathcal{H}} + \alpha \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [-\log \pi_\phi(a | s_i)]. \quad (15)$$

Networks illustrations are available in Appendix D. Note that the bar notation stands for the lagged network with non-trainable parameters. STAC is summarized in Algorithm 1 with gradient descent but Adam optimizer (Kingma & Ba, 2014) is used in our experiments.

Algorithm 1 Stochastic Actor-Critic

Require: Environment `env`

Require: Experience buffer $\mathcal{D}$

Require: Critic $\mathcal{Q}_\theta$, lagged critic $\mathcal{Q}_{\bar{\theta}}$, actor π_ϕ , all with dropout

Require: Initial temperature α , target entropy $\bar{\mathcal{H}}$

Require: Pessimism β

Require: Learning rates $\eta_Q, \eta_\pi, \eta_\alpha$, Polyak parameter ρ

Require: Total training steps N , batch size N_b

```

 $s \sim \text{env.reset}()$  ▷ Reset the environment
for  $N$  timesteps do
   $a \sim \pi_\phi(\cdot | s)$  ▷ Sample action
   $r, s', \text{done} \sim \text{env.step}(a)$  ▷ Act on environment
   $\mathcal{D} \leftarrow \mathcal{D} \cup (s, a, r, s', \text{done})$  ▷ Record transition tuple
  if  $\text{done}$  then  $s \leftarrow s'$  else  $s \sim \text{env.reset}()$  ▷ State transition or reset
  for  $G$  gradient steps do
     $\mathcal{D}_b = \{(s_i, a_i, r_i, s'_i, \text{done}_i)\}_{i=1}^{N_b} \sim \mathcal{D}$  ▷ Sample minibatch for training
     $\tilde{a}'_i \sim \pi_\phi(\cdot | s'_i)$  ▷ Sample next actions
     $Q_i^{TD} = r_i + \gamma(\mu_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \beta\sigma_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \alpha \log \pi_\phi(\tilde{a}'_i | s'_i))(\neg \text{done}_i)$  ▷ Build TD targets
     $\theta \leftarrow \theta - \eta_Q \nabla_\theta \left( \frac{1}{N_b} \sum_{i=1}^{N_b} -\log \mathcal{Q}_\theta(Q_i^{TD} | s_i, a_i) \right)$  ▷ Update critic
     $\phi \leftarrow \phi - \eta_\pi \nabla_\phi \left( \frac{1}{N_b} \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [\mu_\theta(s_i, a) - \beta\sigma_\theta(s_i, a) - \alpha \log \pi_\phi(a | s_i)] \right)$  ▷ Update actor
     $\alpha \leftarrow \alpha - \eta_\alpha \nabla_\alpha \left( -\alpha\bar{\mathcal{H}} + \alpha \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [-\log \pi_\phi(a | s_i)] \right)$  ▷ Update temperature
     $\bar{\theta} \leftarrow \rho\bar{\theta} + (1 - \rho)\theta$  ▷ Update target critic network
  end for
end for

```

5 Experiments

Using the Gymnasium API (Towers et al., 2023), six MuJoCo and three Box2D environments are used for evaluation, as they are standard benchmarks in the literature. From Box2D, **BipedalWalker-v3** and **BipedalWalkerHardcore-v3** model a two-legged robot on randomly generated terrain, while **LunarLander-v3** models a landing spaceship under wind and turbulence, which is maximized in our experiments. Therefore, these environments are inherently stochastic unlike MuJoCo environments.

For evaluation, after each 1000 time steps, we execute a single test episode using the online policy and measure its performance by calculating the total reward accumulated during the episode. In a test episode, dropout and policy temperature are set to zero, following the best practice in the literature. Specified environments are trained through a fixed number of environment interactions, repeated with 5 seeds to assess the stability of the algorithm. Further experimental details are presented in Appendix C.

Hyper-parameters per environment can be found in Table 3 of Appendix C. For all experiments, PyTorch (version 2.7.1) (Paszke et al., 2019) is used. Please refer to Appendix E for the codebase.

5.1 Comparison to Other Algorithms

Our experiments aim to investigate whether enhancing off-policy actor-critic methodology with STAC can improve their sample and computation efficiency on continuous-control benchmarks. For this purpose, STAC is compared to similar competitive algorithms; Distributional Soft-Actor Critic (DSAC) (Ma et al., 2025), and Soft Actor-Critic (SAC) (Haarnoja et al., 2018). Both methods use minimum of double critics for updates. All algorithm results are obtained using in-house code with the same network architectures (including layer normalization but not dropout) to make a fair comparison.

We also add another algorithm called Epistemic Stochastic Actor-Critic (ESTAC), with double distributional critic with one-step uncertainty similar to STAC. For overestimation, minimum of two critic is used for learning similar to SAC and DSAC, instead of temporal aleatoric pessimism. The purpose is to isolate the source of STAC’s improvements by replacing temporal aleatoric pessimism with epistemic overestimation mitigation.

The pessimism rates and dropout configurations used for comparison are resulted from ablation studies, and explained in the next section. Optimal configurations are summarized in Table 4.

Learning curves In Figure 2, the performance of STAC is shown against previously mentioned state-of-the-art algorithms. Additionally, value estimation errors are presented in Figure 3. Value estimation error is measured as the difference between the critic’s predicted action value and the discounted return collected during evaluation episodes. The bold lines represent the inter-quartile mean across random seeds, while the shaded area indicates the corresponding quartile ranges of the total reward across different seeds. Curves are smoothed through time for better visibility.

Final performane of learning processes averaged over random seeds are summarized in Table 1.

Sample efficiency As seen from Figure 2 and Table 1, STAC outperforms other algorithms on some environments in terms of sample efficiency (see **BipedalWalkerHardcore-v3**, **Hopper-v4**, **Swimmer-v4**, **Walker2d-v4**) while it fall behind in other environments. However, it is important to note that there are many factors that affect performance, such as distribution modeling method, number of critics and regularization method, network size and learning rates etc. Rather than absolute performance alone, we demonstrate that STAC achieves competitive or superior performance without convergence issues while using a single critic, whereas competing methods rely on double critics.

ESTAC vs STAC Except on **BipedalWalker-v3**, STAC consistently outperforms ESTAC across all environments. This indicates that temporal aleatoric pessimism is generally more effective than epistemic pessimism, at least in our experimental setting.

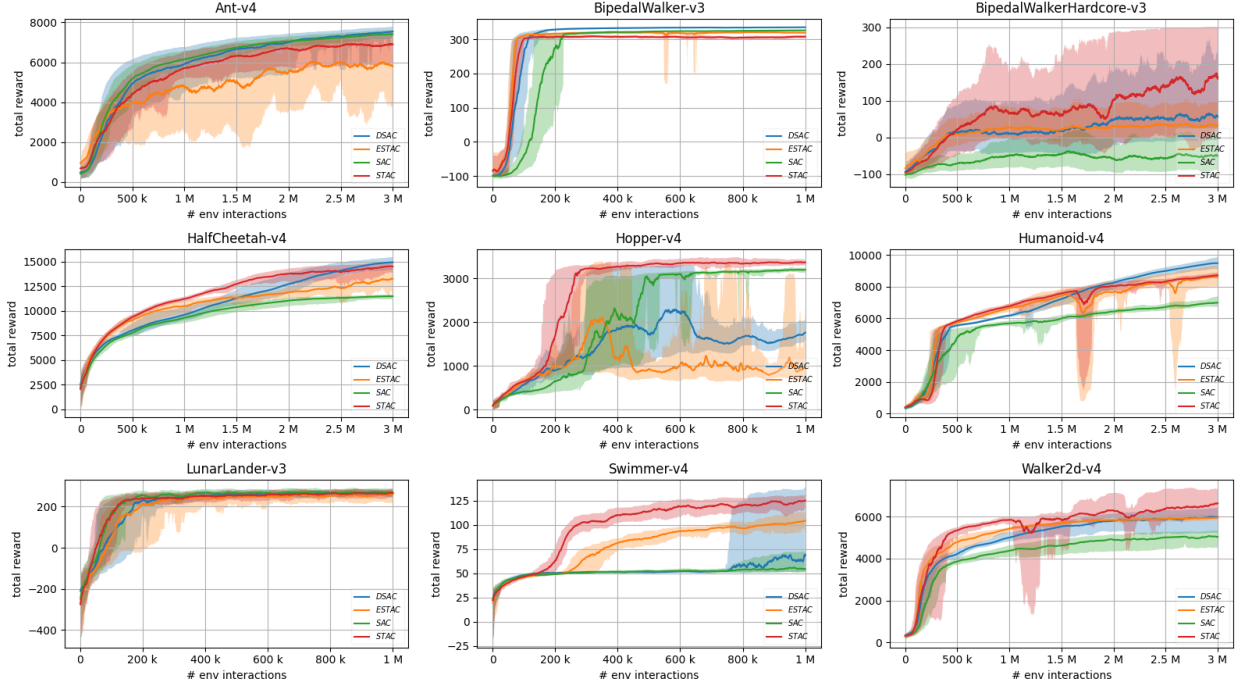


Figure 2: Episodic score curves of STAC and other algorithms.

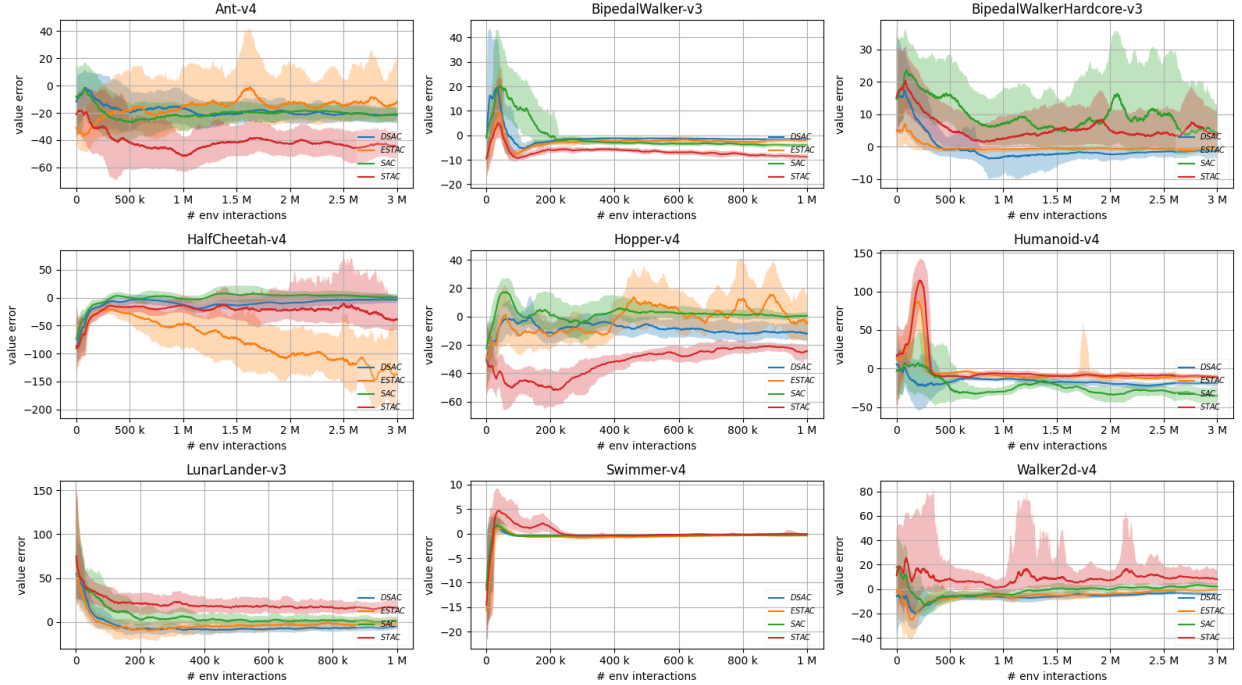


Figure 3: Average episodic value estimation error of STAC and other algorithms.

Table 1: Inter-quartile mean results of last %1 of evaluation episodes, β and dropout for STAC is selected with best performance.

Env	# steps	DSAC	ESTAC	SAC	STAC
Ant-v4	3M	7543.1	5798.9	7407.1	6907.3
BipedalWalker-v3	1M	335.3	319.2	326.3	308.5
BipedalWalkerHardcore-v3	3M	56.8	32.3	-50.8	159.9
HalfCheetah-v4	3M	14921.5	13241.6	11475.6	14569.3
Hopper-v4	1M	1762.9	944.3	3196.4	3363.7
Humanoid-v4	3M	9484.1	8608.9	6990.4	8726.2
LunarLander-v3	1M	265.4	258.3	268.9	265.7
Swimmer-v4	1M	69.3	104.1	54.2	125.0
Walker2d-v4	3M	5981.4	5952.5	5036.4	6634.7

5.2 Sweeping Pessimism and Dropout

STAC is experimented under varying levels of pessimism β and dropout rates to isolate effect of pessimism and dropout individually. We used 5 pessimism level and 4 dropout configurations: no dropout, actor dropout, critic dropout and actor&critic dropout.

Table 2: Inter-quartile mean scores of last %1 of evaluation episodes, for varying β . Dropout rate is 0.01 for critic and actor.

Env	$\beta = 0.0$	$\beta = 0.125$	$\beta = 0.25$	$\beta = 0.375$	$\beta = 0.5$
Ant-v4	5264.4	7082.7	6907.3	6246.4	4834.4
BipedalWalker-v3	171.9	125.4	262.6	308.0	305.1
BipedalWalkerHardcore-v3	61.8	118.8	43.4	-0.0	-59.8
HalfCheetah-v4	14489.5	12299.6	13385.4	11042.1	7602.7
Hopper-v4	1132.4	1308.8	2643.4	3318.2	3363.7
Humanoid-v4	5623.9	8407.2	8277.9	7587.2	6930.8
LunarLander-v3	265.7	263.8	157.2	14.3	-77.7
Swimmer-v4	103.7	93.5	70.0	54.2	45.9
Walker2d-v4	4299.2	6306.4	5842.1	5795.2	5539.1

Learning curves In the experiments, we presented controlled study to isolate dropout’s role: while distributional pessimism eliminates overestimation bias, dropout primarily increases training stability and performance. Both learning curves are available in Appendix B. Episodic scores and related value estimation errors for varying level of pessimism under joint actor/critic dropout are summarized in Figure 5 and Figure 6. In addition, same curves for different dropout configurations under same level of pessimism are summarized in Figure 7 and Figure 8, where β values are in Table 4. The shaded area represents quartile limits, while the solid line represents inter-quartile mean across different seeds and %1 smoothing window of evaluation episodes through environment steps.

Pessimism Table 2 summarizes final performance in terms of inter-quartile mean score on last %1 evaluation episodes. Results indicate that β is a sensitive parameter, the higher β yields a negative error (see Figure 6), consistent with the intended effect. In addition, score curves are worse if value estimations tend to be positive, meaning that critic overestimation is not mitigated enough (see Ant-v4, BipedalWalker-v3, Hopper-v4, Humanoid-v4, Walker-v4). On the other hand, score curves are again worse when error curves are negative than it should be, meaning that the agent is stuck on critic underestimation caused by high pessimism (see BipedalWalkerHardcore-v3, HalfCheetah-v4, LunarLander-v3, Swimmer-v4). In the end, this sensitivity varies for different environments, possibly due to differences in reward sparsity and stochasticity. This parameter stands as the major bottleneck of STAC and can only be determined by this heuristic for now.

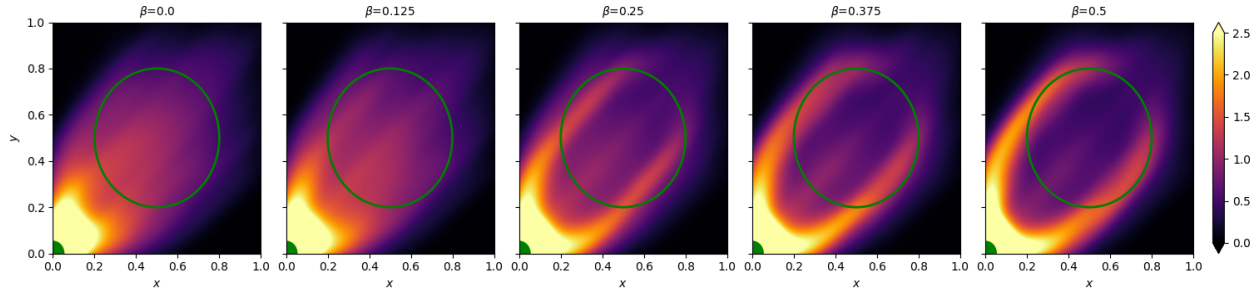


Figure 4: Position occurrence density heatmap of STAC on RiskyPointMass-v0 for different β values. The green circle on the middle represents boundary of danger zone. Episodes are terminated when point mass enters green area on the lower left.

Dropout Dropout’s primary role in STAC is as a regularization tool to improve optimization stability and robustness to noisy data. In Figure 7, independent of critic dropout, actor dropout consistently improves performance by stabilizing policy updates. According to Figure 8, the overestimation trends observed with and without dropout are qualitatively similar, confirming that our core contribution: (mitigating overestimation via aleatoric uncertainty) remains effective independent of dropout, except **Hopper-v4**. However, in this environment, learning does not even converge without critic dropout. Therefore, it is necessary in some environments for training stability rather than overestimation mitigation. Double/ensemble critic approaches mitigate training instability naturally, whereas STAC requires a convenient dropout in some cases.

Pessimism under Environment Stochasticity Pessimistic learning based on aleatoric uncertainty can interact with exploration, especially in environments with complex dynamics. Comparing **BipedalWalker-v3** and **BipedalWalkerHardcore-v3**, we observe that lower pessimism performs better in the harder environment, while moderate pessimism improves performance in the simpler setting (see Figure 5 and 6). One possible explanation is that, in simpler tasks, uncertainty is dominated by approximation and bootstrapping effects, where pessimism helps reduce overestimation. In contrast, in more challenging environments, uncertainty may be driven more strongly by the environment itself, and excessive pessimism can slow learning. These observations suggest that the appropriate level of pessimism depends on the interaction between task complexity and learning dynamics.

5.3 Risk Sensitivity by Pessimism

In STAC, actor updates are pessimistic for one-step return uncertainty. However, critic targets are also pessimistic with respect to consecutive steps. Therefore, pessimism is also bootstrapped along with reward, and increasing risk-averse behavior is expected as β increases. For this purpose, a toy problem is used to demonstrate this behavior.

Environment Details Following Ma et al. (2021) and Ma et al. (2025), risky navigation task is used as a simple stochastic problem. The environment is named as **RiskyPointMass-v0**. The goal of this task is to navigate a point mass towards a target point while avoiding a danger zone, in a two dimensional space. Danger zone is a circular area with a radius of $r_0 = 0.3$, centered at point $(0.5, 0.5)$. Initial states are randomly selected from $\mathcal{U}(0.3, 1)$ for both coordinates, excluding the danger zone. Episode terminates if the agent is close to target point $(0, 0)$ by Euclidean distance of 0.05. At each step, the agent receives the sum of three reward components: negative Euclidean distance to the target, fixed -0.1 points to promote episode termination, a penalty of -10 points if the agent is within danger zone with probability $0.1 \exp(-4r^2/r_0^2)$ where r is the distance to danger zone center. The agent can move by 0.1 unit at most in both dimensions plus a random slip (at most 0.02 unit). The reward function is designed so that optimal risk-neutral solution is to navigate linearly to the target point whereas optimal risk-averse solution is to navigate around the danger zone circle.

Results For each β value, STAC is run with a single fixed seed, as this experiment is intended to provide a qualitative illustration of risk-sensitive behavior rather than a performance comparison. In Figure 4, for varying β values, position densities of the trained agents are demonstrated. The bigger green circles represent danger zone boundaries, whereas quarter green circles on the lower left represent the target points. For this demonstration, 500 evaluation steps are run with random initial points. Clearly, with higher β values, the agent prefers to go around the circle as opposed to agents trained with lower β . This is an expected effect since pessimism is bootstrapped through TD targets, effectively inducing conservative behavior over the roll-out. To summarize, STAC not only mitigates overestimation but also induces risk-averse learning with pessimism on temporal uncertainty.

6 Related Work

Overestimation reduction by epistemic uncertainty In the literature, overestimation problem is solved by pessimistic learning based on *epistemic uncertainty* of critic (Chen et al., 2021; Hiraoka et al., 2021). The same approach is also used in model-based RL methods (Janner et al., 2019; Chua et al., 2018; Depeweg et al., 2016). Recently, for off-policy model-free actor-critic algorithms, epistemic uncertainty is estimated by using double networks (Fujimoto et al., 2018; Haarnoja et al., 2018), or ensemble networks (Chen et al., 2021; Moskovitz et al., 2021), or dropout (Hiraoka et al., 2021). Ensemble of critics are computationally expensive as there are multiple of parameters to be optimized. The mentioned methods use a constant level of pessimism for policy evaluation and improvement, while Moskovitz et al. (2021) focused on updating pessimism *on the fly* as a bandit problem rather than fixing it. Similarly Cetin & Celiktutan (2023) tunes pessimism online by treating it as a dual variable by enforcing the expected off-policy action-value bias to be zero.

Risk-sensitive reinforcement learning Aleatoric uncertainty representation for the value function carries fundamental importance, especially in the presence of approximation (Bellemare et al., 2017). This can be conducted by atoms (Bellemare et al., 2017), quantiles (Dabney et al., 2018b; Ma et al., 2025), or a normal distribution (Tang et al., 2019; Yang et al., 2021) to model cumulative return behavior. For safety-critical RL applications to avoid catastrophic situations, aleatoric pessimism is used by Tang et al. (2019) and Yang et al. (2021). Different than STAC, these methods use pessimism only on actor update for safety, not for overestimation reduction, and models full return uncertainty.

Overestimation reduction by aleatoric uncertainty Kuznetsov et al. (2020) claims *aleatoric uncertainty* is also responsible for overestimation since any randomness is exploited when the Bellman optimality operator ($\mathcal{T}^*$) is employed. For this purpose, they use ensemble networks for epistemic uncertainty, in which each network is a distributional network (as quantiles) for aleatoric uncertainty assessment, and used both types of uncertainties for overestimation correction. Similarly, STAC uses distributional critic, but only models temporal (one-step) aleatoric uncertainty rather than full return distribution as this is the main source of overestimation, without epistemic modeling.

7 Conclusion & Future Directions

In this paper, we introduced Stochastic Actor-Critic (STAC), a novel off-policy actor-critic algorithm. The main idea is to mitigate overestimation for the sake of faster and more robust learning by incorporating the pessimistic learning objective using temporal aleatoric critic uncertainty. For this purpose, the critic is modeled as a distributional neural network. Although normal distribution is used in STAC, our analysis is valid for all sub-Gaussian critic representations.

We derived an upper bound for overestimation, demonstrating that an adequate level of pessimism mitigates overestimation without succumbing to underestimation, thus facilitating computation and sample-efficient learning. As a by-product, this approach also yields risk-averse learning process. Lastly, dropout on the actor and critic networks is proposed to mitigate overfitting.. Actor dropout always improves performance, whereas using dropout on critic is only required to ensure learning convergence.

Adaptive Pessimism and Regularization While STAC demonstrates that temporal aleatoric pessimism alone is sufficient to mitigate overestimation using a single critic, several directions remain open. In particular, the pessimism coefficient β is currently selected empirically and its optimal value varies across environments, motivating future work on adaptive or principled tuning strategies, similar to Moskovitz et al. (2021); Cetin & Celiktutan (2023). In addition, although STAC reduces computational overhead by avoiding double critics, a single-critic setup may require explicit regularization (e.g., dropout) for stable optimization in some environments. For future work, alternative regularization approaches might be considered for more stable training, including ensembles.

Distribution Modeling The effect of pessimism on highly stochastic environments is also an important topic for research. As shown, pessimistic updates lead to risk-averse behavior. Our results are limited to simple stochastic tasks, and it is worth investigating STAC empirically on more difficult ones. For more advanced modeling, aleatoric critic uncertainty can be modeled by quantiles (Dabney et al., 2018b; Ma et al., 2025) or flow matching (Chen et al., 2025; Dong et al., 2025), rather than normal distribution.

Exploration Lastly, current methods use epistemic critic uncertainty for overestimation problem, which contradicts with *optimism in the face of uncertainty* principle. Future researches might consider using epistemic critic and actor uncertainty for exploration, retaining temporal aleatoric pessimism.

Broader Impact STAC is an important contribution to understand overestimation phenomenon in off-policy actor-critic learning. It demonstrates that critic overestimation is also a topic of distributional learning, and the solution is to devise a risk-averse objective for both actor and critic. Moreover, it also proves that modeling epistemic uncertainty is not necessary at all for this problem, improving computation efficiency directly.

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References

- Mastane Achab. *Ranking and risk-aware reinforcement learning*. PhD thesis, Institut polytechnique de Paris, 2020.
- Mastane Achab, Reda Alami, Yasser Abdelaziz Dahou Djilali, Kirill Fedyanin, and Eric Moulines. One-step distributional reinforcement learning. *arXiv preprint arXiv:2304.14421*, 2023.
- Jean-Yves Audibert, Rémi Munos, and Csaba Szepesvári. Tuning bandit algorithms in stochastic environments. In *International conference on algorithmic learning theory*, pp. 150–165. Springer, 2007.
- Kamyar Azizzadenesheli, Emma Brunskill, and Animashree Anandkumar. Efficient exploration through bayesian deep q-networks. In *2018 Information Theory and Applications Workshop (ITA)*, pp. 1–9. IEEE, 2018.
- Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- Marc G Bellemare, Will Dabney, and Rémi Munos. A distributional perspective on reinforcement learning. In *International conference on machine learning*, pp. 449–458. PMLR, 2017.
- Edoardo Cetin and Oya Celiktutan. Learning pessimism for reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pp. 6971–6979, 2023.
- Deshu Chen, Yuchen Liu, Zhijian Zhou, Chao Qu, and Yuan Qi. Unleashing flow policies with distributional critics. *arXiv preprint arXiv:2509.23087*, 2025.
- Xinyue Chen, Che Wang, Zijian Zhou, and Keith Ross. Randomized ensembled double q-learning: Learning fast without a model. *arXiv preprint arXiv:2101.05982*, 2021.

-
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. *Advances in neural information processing systems*, 31, 2018.
- Kamil Ciosek, Quan Vuong, Robert Loftin, and Katja Hofmann. Better exploration with optimistic actor critic. *Advances in Neural Information Processing Systems*, 32, 2019.
- Will Dabney, Georg Ostrovski, David Silver, and Rémi Munos. Implicit quantile networks for distributional reinforcement learning. In *International conference on machine learning*, pp. 1096–1105. PMLR, 2018a.
- Will Dabney, Mark Rowland, Marc Bellemare, and Rémi Munos. Distributional reinforcement learning with quantile regression. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018b.
- Stefan Depeweg, José Miguel Hernández-Lobato, Finale Doshi-Velez, and Steffen Udluft. Learning and policy search in stochastic dynamical systems with bayesian neural networks. *arXiv preprint arXiv:1605.07127*, 2016.
- Armen Der Kiureghian and Ove Ditlevsen. Aleatory or epistemic? does it matter? *Structural safety*, 31(2): 105–112, 2009.
- Perry Dong, Chongyi Zheng, Chelsea Finn, Dorsa Sadigh, and Benjamin Eysenbach. Value flows. *arXiv preprint arXiv:2510.07650*, 2025.
- Jingliang Duan, Yang Guan, Shengbo Eben Li, Yangang Ren, Qi Sun, and Bo Cheng. Distributional soft actor-critic: Off-policy reinforcement learning for addressing value estimation errors. *IEEE transactions on neural networks and learning systems*, 33(11):6584–6598, 2021.
- Jingliang Duan, Wenxuan Wang, Liming Xiao, Jiaxin Gao, Shengbo Eben Li, Chang Liu, Ya-Qin Zhang, Bo Cheng, and Keqiang Li. Distributional soft actor-critic with three refinements. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- Scott Fujimoto, Herke Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International conference on machine learning*, pp. 1587–1596. PMLR, 2018.
- Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML-16)*, 2016.
- Yarin Gal, Rowan McAllister, and Carl E. Rasmussen. Improving PILCO with Bayesian neural network dynamics models. In *Data-Efficient Machine Learning workshop, ICML*, April 2016a.
- Yarin Gal, Jiri Hron, and Alex Kendall. Concrete dropout. *Advances in neural information processing systems*, 30, 2017.
- Yarin Gal et al. Uncertainty in deep learning. 2016b.
- Tuomas Haarnoja, Haoran Tang, Pieter Abbeel, and Sergey Levine. Reinforcement learning with deep energy-based policies. In *International conference on machine learning*, pp. 1352–1361. PMLR, 2017.
- Tuomas Haarnoja, Aurick Zhou, Kristian Hartikainen, George Tucker, Sehoon Ha, Jie Tan, Vikash Kumar, Henry Zhu, Abhishek Gupta, Pieter Abbeel, et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018.
- Qiang He, Huangyuan Su, Chen Gong, and Xinwen Hou. Mepg: A minimalist ensemble policy gradient framework for deep reinforcement learning. *arXiv preprint arXiv:2109.10552*, 2021.
- Takuya Hiraoka, Takahisa Imagawa, Taisei Hashimoto, Takashi Onishi, and Yoshimasa Tsuruoka. Dropout q-functions for doubly efficient reinforcement learning. *arXiv preprint arXiv:2110.02034*, 2021.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. *Advances in neural information processing systems*, 32, 2019.

-
- Natasha Jaques, Asma Ghandeharioun, Judy Hanwen Shen, Craig Ferguson, Agata Lapedriza, Noah Jones, Shixiang Gu, and Rosalind Picard. Way off-policy batch deep reinforcement learning of implicit human preferences in dialog. *arXiv preprint arXiv:1907.00456*, 2019.
- Gregory Kahn, Adam Villafior, Vitchyr Pong, Pieter Abbeel, and Sergey Levine. Uncertainty-aware reinforcement learning for collision avoidance. *arXiv preprint arXiv:1702.01182*, 2017.
- Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? *Advances in neural information processing systems*, 30, 2017.
- Dohyeong Kim and Songhwa Oh. Efficient off-policy safe reinforcement learning using trust region conditional value at risk. *IEEE Robotics and Automation Letters*, 7(3):7644–7651, 2022.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Levente Kocsis and Csaba Szepesvári. Bandit based monte-carlo planning. In *European conference on machine learning*, pp. 282–293. Springer, 2006.
- Arsenii Kuznetsov, Pavel Shvechikov, Alexander Grishin, and Dmitry Vetrov. Controlling overestimation bias with truncated mixture of continuous distributional quantile critics. In *International Conference on Machine Learning*, pp. 5556–5566. PMLR, 2020.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017.
- Qingfeng Lan, Yangchen Pan, Alona Fyshe, and Martha White. Maxmin q-learning: Controlling the estimation bias of q-learning. *arXiv preprint arXiv:2002.06487*, 2020.
- Sicen Li, Qinyun Tang, Yiming Pang, Xinmeng Ma, and Gang Wang. Realistic actor-critic: A framework for balance between value overestimation and underestimation. *Frontiers in Neurorobotics*, 16:1081242, 2023.
- Timothy P Lillicrap, Jonathan J Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015.
- Xiaoteng Ma, Junyao Chen, Li Xia, Jun Yang, Qianchuan Zhao, and Zhengyuan Zhou. Dsac: Distributional soft actor-critic for risk-sensitive reinforcement learning. *Journal of Artificial Intelligence Research*, 83, 2025.
- Yecheng Ma, Dinesh Jayaraman, and Osbert Bastani. Conservative offline distributional reinforcement learning. *Advances in neural information processing systems*, 34:19235–19247, 2021.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Alex Graves, Ioannis Antonoglou, Daan Wierstra, and Martin Riedmiller. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.
- Thomas M Moerland, Joost Broekens, and Catholijn M Jonker. Efficient exploration with double uncertain value networks. *arXiv preprint arXiv:1711.10789*, 2017.
- Ted Moskovitz, Jack Parker-Holder, Aldo Pacchiano, Michael Arbel, and Michael Jordan. Tactical optimism and pessimism for deep reinforcement learning. *Advances in Neural Information Processing Systems*, 34: 12849–12863, 2021.
- Michal Nauman, Michał Bortkiewicz, Piotr Miłoś, Tomasz Trzciński, Mateusz Ostaszewski, and Marek Cygan. Overestimation, overfitting, and plasticity in actor-critic: the bitter lesson of reinforcement learning. *arXiv preprint arXiv:2403.00514*, 2024.
- Brendan O’Donoghue. Efficient exploration via epistemic-risk-seeking policy optimization. *arXiv preprint arXiv:2302.09339*, 2023.

-
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019.
- Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research*, 15(1): 1929–1958, 2014.
- Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.
- Yichuan Charlie Tang, Jian Zhang, and Ruslan Salakhutdinov. Worst cases policy gradients. *arXiv preprint arXiv:1911.03618*, 2019.
- Thibaut Théate and Damien Ernst. Risk-sensitive policy with distributional reinforcement learning. *Algorithms*, 16(7):325, 2023.
- Sebastian Thrun and Anton Schwartz. Issues in using function approximation for reinforcement learning. In *Proceedings of the 1993 connectionist models summer school*, pp. 255–263. Psychology Press, 2014.
- Mark Towers, Jordan K. Terry, Ariel Kwiatkowski, John U. Balis, Gianluca de Cola, Tristan Deleu, Manuel Goulão, Andreas Kallinteris, Arjun KG, Markus Krimmel, Rodrigo Perez-Vicente, Andrea Pierré, Sander Schulhoff, Jun Jet Tai, Andrew Tan Jin Shen, and Omar G. Younis. Gymnasium, March 2023. URL <https://zenodo.org/record/8127025>.
- Hado Van Hasselt, Yotam Doron, Florian Strub, Matteo Hessel, Nicolas Sonnerat, and Joseph Modayil. Deep reinforcement learning and the deadly triad. *arXiv preprint arXiv:1812.02648*, 2018.
- Xinyang Wu, Mohamed El-Shamouty, Christof Nitsche, and Marco F Huber. Uncertainty-guided active reinforcement learning with bayesian neural networks. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5751–5757. IEEE, 2023.
- Qisong Yang, Thiago D Simão, Simon H Tindemans, and Matthijs TJ Spaan. Wcsac: Worst-case soft actor critic for safety-constrained reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 10639–10646, 2021.
- Junwei Zhang, Shuai Han, Xi Xiong, Sheng Zhu, and Shuai Lü. Explorer-actor-critic: Better actors for deep reinforcement learning. *Information Sciences*, 662:120255, 2024.

Appendix A Proofs

Proof of Theorem 3.1. Analyzing Bellman update $\mathcal{T}^*Q(s, a)$,

$$\begin{aligned}
\mathcal{T}^*Q(s, a) &= R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\mathbb{E}_{Q \sim \mathcal{Q}} \left[\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right] \right) \\
&= R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \mathbb{E}_{Q \sim \mathcal{Q}} \left[\exp(\tilde{\alpha}^{-1} Q(s', a')) \right] da' \right) \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right)
\end{aligned}$$

First inequality is by Jensen's inequality (using concave property of log function) while the following equality is a result of Tonelli's theorem. The second inequality results from the sub-Gaussian assumption 3.1. $\square$

Proof of Corollary 3.1.1. From the Theorem 3.1, we can show that

$$\begin{aligned}
\mathcal{T}_\beta^*Q(s, a) &= R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a') - \beta \sigma(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} (\mu(s', a') - \beta \sigma(s', a') + \frac{1}{2} \tilde{\alpha}^{-1} \sigma^2(s', a'))) da' \right) \\
&= R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu^\dagger(s', a')) da' \right) = \mathcal{T}^* \mu^\dagger(s, a).
\end{aligned}$$

where we have defined $\mu^\dagger(s', a') = \mu(s', a') - \beta \sigma(s', a') + \frac{1}{2} \tilde{\alpha}^{-1} \sigma^2(s', a')$. If $\beta \geq \max_{(s', a')} \frac{1}{2} \tilde{\alpha}^{-1} \sigma(s', a')$, then $\mu^\dagger(s', a') < \mu(s', a')$. So we can show that

$$\mathcal{T}_\beta^*Q(s, a) \leq \mathcal{T}^* \mu^\dagger(s, a) \leq \mathcal{T}^* \mu(s, a). \quad (16)$$

$\square$

Proof of Theorem 3.2. From the Theorem 3.1, we can show that

$$\begin{aligned}
\mathbb{E}_{s' \sim \tau} [\mathcal{T}^*Q(s, a)] &\leq R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a')) da' \right) \cdot \left(\max_{a'} \exp(\frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) \right) \right) \right] \\
&= R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a')) da' \right) \right] + \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right].
\end{aligned}$$

The second inequality is a result of the mean value theorem for integrals. In the last equality, the first two terms are equal to $\mathcal{T}^* \mu(s, a)$. Therefore,

$$\epsilon(s, a) = \mathbb{E}_{s' \sim \tau} [\mathcal{T}^*Q(s, a)] - \mathcal{T}^* \mu(s, a) \leq \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right].$$

$\square$

Appendix B Results of Ablation Studies

B.1 Pessimism Ablation

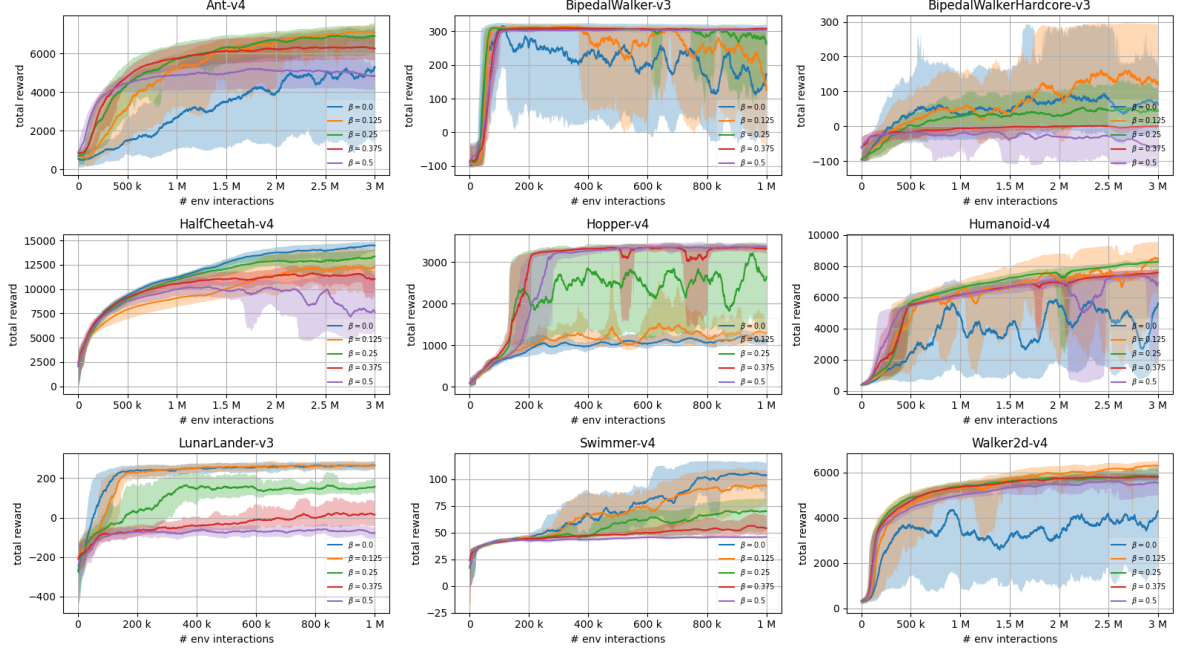


Figure 5: Episodic score curves of STAC with varying pessimism (β) parameter, with actor and critic dropout equal to 0.01.

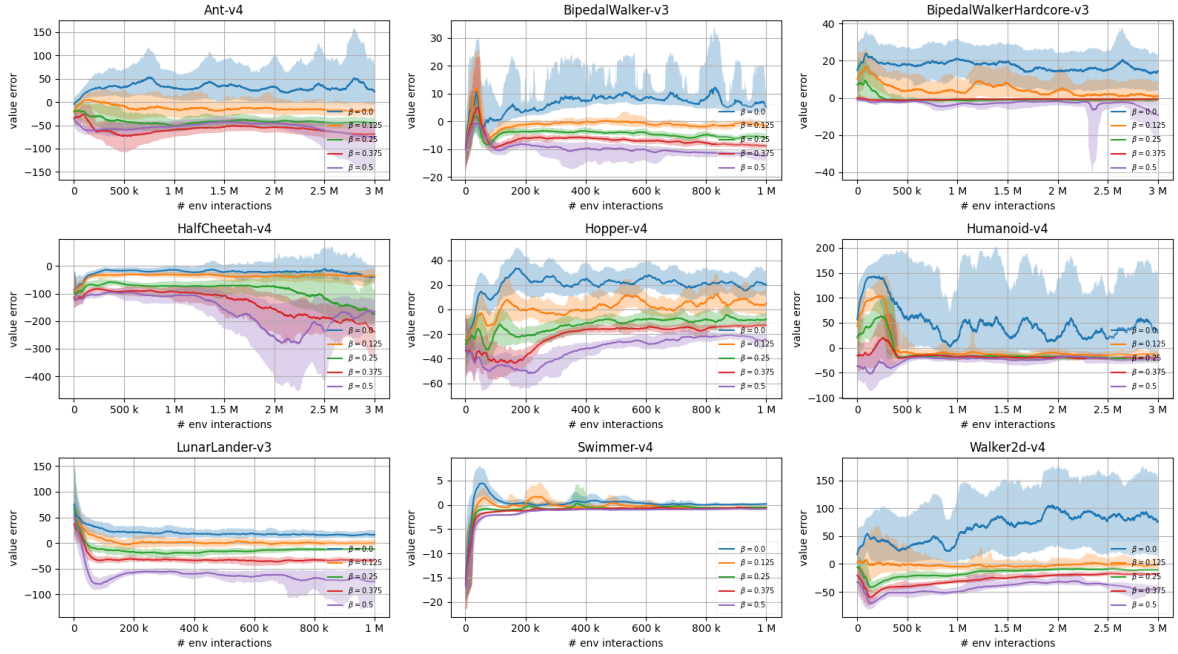


Figure 6: Average episodic value estimation error curves of STAC with varying pessimism (β) parameter, with actor and critic dropout equal to 0.01.

B.2 Dropout Configuration Ablation

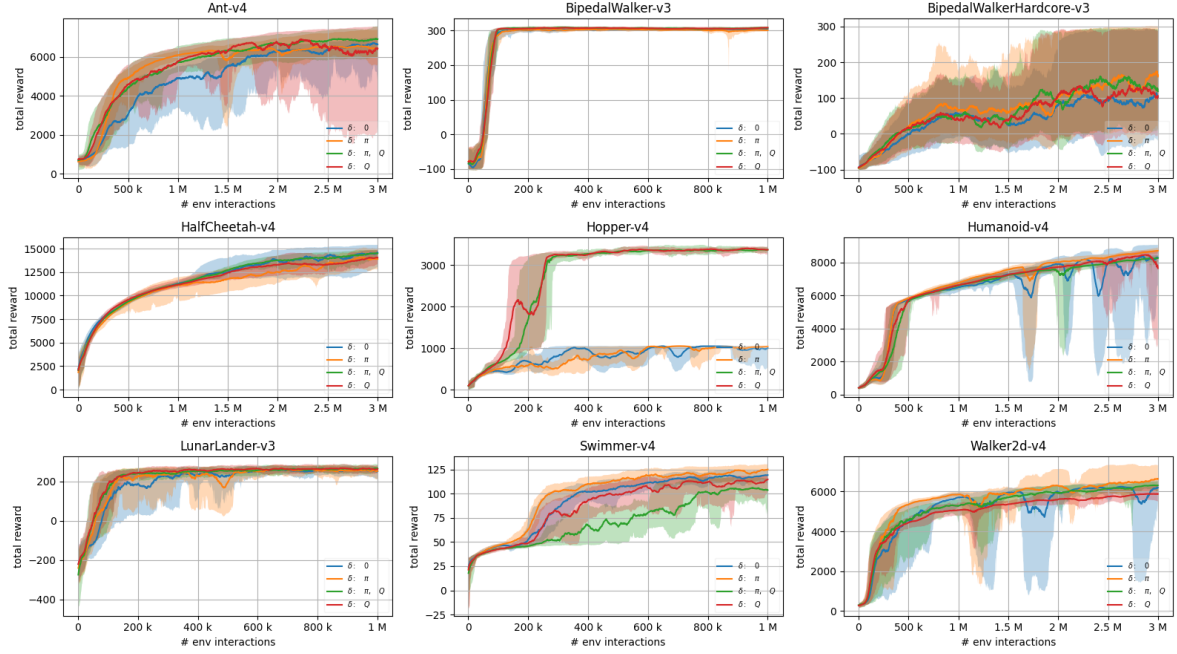


Figure 7: Episodic score curves of STAC with 4 different dropout configurations, under fixed pessimism level yielding best performance (see Table 4).

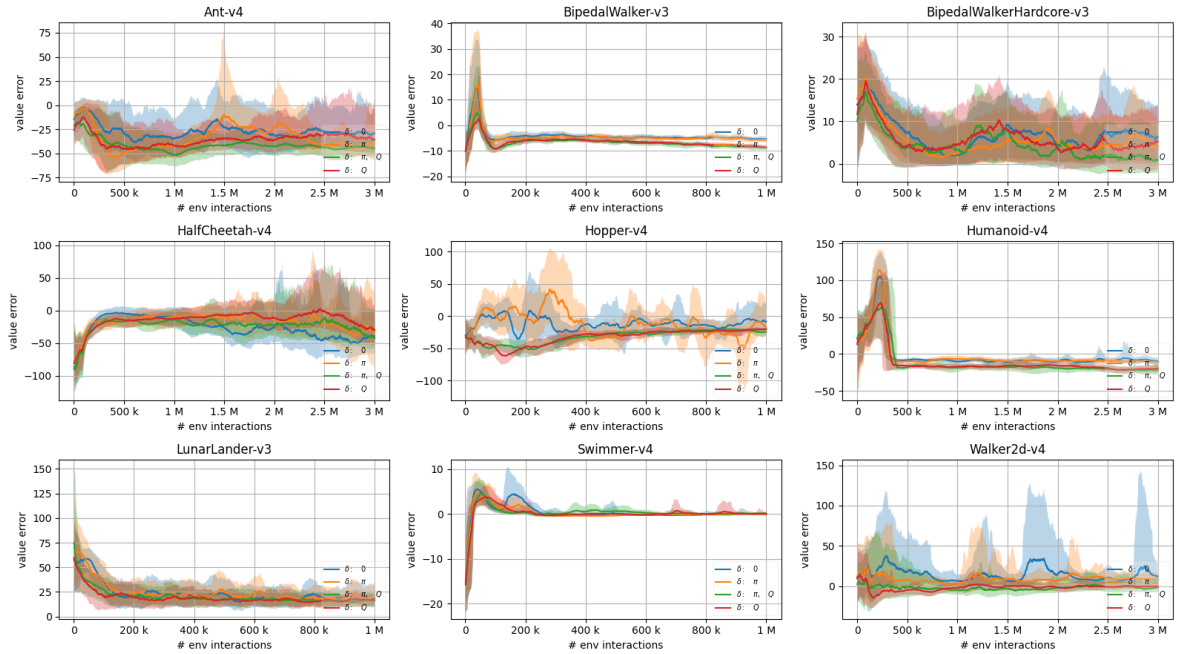


Figure 8: Average episodic value estimation error curves of STAC with 4 different dropout configurations, under fixed pessimism level yielding best performance (see Table 4).

Appendix C Hyper-parameters and Experiment Details

Table 3: Experimental Parameters per Algorithm

Algorithm	Parameter	Value
DSAC, ESTAC, SAC, STAC	Optimizer	Adam ((Kingma & Ba, 2014))
	Critic Learning Rate	3×10^{-4}
	Actor Learning Rate	3×10^{-4}
	Discount Rate (γ)	0.99
	Target-Smoothing Coefficient (ρ)	0.995
	Target Entropy ($\bar{H}$)	$- \mathcal{A} $
	Replay Buffer Size	1×10^6
	Mini-Batch Size	256
	Learning Starting Step	10000
	UTD Ratio (G)	1
DSAC	Number of Quantiles	25
	Huber Loss threshold (κ)	1.0

Table 4: Target policy entropy ($\bar{H}$), pessimism β and dropout rates per environment, yielding best results. All algorithms share same policy entropy.

Environment	Entropy ($\bar{H}$)	Pessimism (β)	Actor Dropout	Critic Dropout
Ant-v4	-4	0.25	0.01	0.01
BipedalWalker-v3	-2	0.375	0.0	0.01
BipedalWalkerHardcore-v3	-2	0.125	0.01	0
HalfCheetah-v4	-3	0.0	0.0	0.0
Hopper-v4	-1	0.5	0.01	0.01
Humanoid-v4	-8	0.25	0.01	0
LunarLander-v3	-2	0.0	0.01	0.01
Swimmer-v4	-1	0.0	0.01	0
Walker2d-v4	-3	0.125	0.01	0

Appendix D Network Architectures of STAC

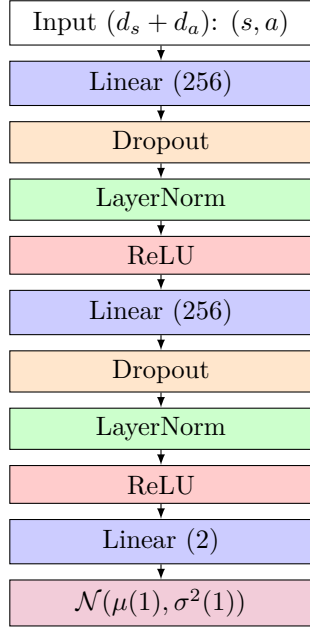


Figure 9: Critic network architecture

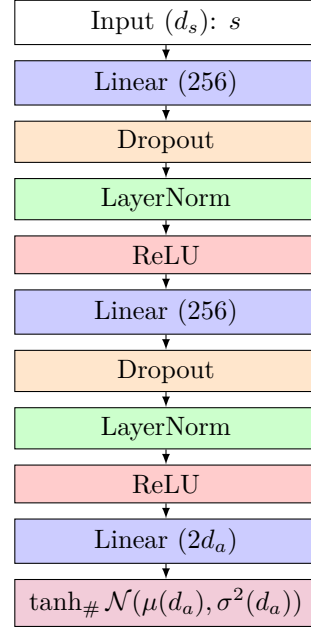


Figure 10: Actor network architecture

Appendix E Source Code

Our results can be accessed publicly at <https://github.com/authors-github/stochastic-actor-critic-results>. This code uses our in-house developed RL framework as a sub-repository, available on <https://github.com/authors-github/rl-warehouse>.